Mastering CSS for Selenium Test Automation

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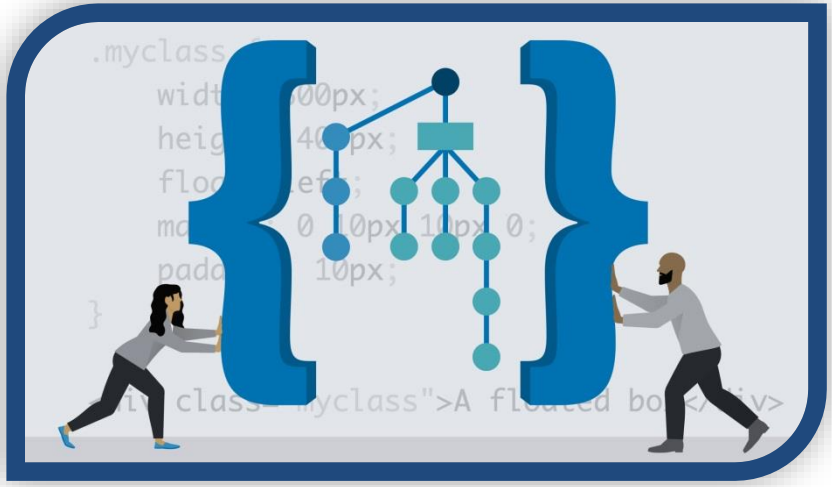
CSS Selectors

CSS Selector is the combination of an element selector and a selector value which identifies the web element within a web page. The composite of an element selector and selector value is known as Selector Pattern. Selector Pattern is constructed using HTML tags, attributes and their values. CSS Selector and XPath are very much similar underlying the only difference in their construction protocol.

Like XPath, CSS selector can also locate web elements having no ID, class or Name.

So now gearing ahead, let us discuss the primitive types of CSS Selectors:



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1. Locating Elements by its Attributes

Syntax : elementName[attributeName='value']

```
<input id="txtUsername" type="text" name="txtUsername">
```

Following are possible CSS

1. Css: input[id='txtUsername']

2. Css: input[name='txtUsername']

3. Css: input[type='text']

Third CSS should not be used even though it is a valid CSS. It is not unique. There could be many elements with attribute type='text'.

2. LocatingElementsbyID

InCSS#signisusedfor locating the elementsbyvalue of ID.

Syntax: elementName#valueofID or #valueofID

```
<input id="txtUsername" type="text" name="txtUsername">
```

Examples :
Css: input#txtUsername
Css: #txtUsernam

3. Locating Elements by Class

Locating element by single class

Syntax :elementName.class or .class

<button class="btnprimary" tabindex="3" type="submit">Sign in</button>

Examples :

Css: button.btnprimary

Css: . btnprimary

Locating elements by multiple classes

Syntax :elementName.class1.class2.class3 or .class1.class2.class3

<button class="submit btn primarybtn " type="submit">
Log in
</button>

Examples :

Css: button.btn.primarybtn

Css: button.submit.btn.primarybtn

Css: .btn.primarybtn

Css: .submit.btn.primarybtn

4. Locating Elements by Class and Attribute

Syntax : elementName.class[attributeName='value']

<button class="submit btn primarybtn flextablebtn jssubmit" type="submit">Log in</button>

Example :

Css: button.primarybtn[type='submit']

5. Locating Elements with more than one Attribute

Syntax :elementName[attribute1='value1'][attribute2='value2']...[attributeN='valueN']

Or

[attribute1='value1'][attribute2='value2']...[attributeN='valueN']

<input id="txtUsername" type="text" name="txtUsername" class='userName' >

Examples : Css: input[id='txtUsername'][name='txtUsername']

6. Locating Child Elements by Element Name

Locating direct child or sub child (child at any level)

Syntax : <CSS of Parent Element><space><elementName>

```
<form id="frmLogin" method="post" action="/index.php/auth/validations">
<div id="divUsername" class="textInputContainer">
<input id="txtUsername" type="text" name="txtUsername">
</div>
.....
</form>
```

Example for locating sub child :

Css: form#frmLogin input

Locating direct child

Syntax : CSSofParent Element > elementName

Example :

Css: form > div > input

7. Locating Nth Child Elements by Element type

Syntax : CSSofParentElement elementName:nthoftype(N)

```
<form id="frmLogin" method="post" action="/index.php/auth/validations">
<input id="txtUsername" type="text" name="txtUsername">
<input name="txtPassword" id="txtPassword" type="password">
<input name="Submit" class="button" id="btnLogin" value="LOGIN" type="submit">
</form>
```

Example :

To locate the password field

Css: form#frmLogin input:nth-of-type(2)

8. Locating First Child Element by Element Name

Syntax : CSSofParentElement elementName:firstchild

```
<form id="frmLogin" method="post" action="/index.php/auth/validations">
<input id="txtUsername" type="text" name="txtUsername">
<input name="txtPassword" id="txtPassword" type="password">
<input name="Submit" class="button" id="btnLogin" value="LOGIN" type="submit">
</form>
```

Example :

To locate the username field

Css: form#frmLogin input:first-child

9. Locating Last Child Element by Element Name

Syntax : <CSS of Parent Element><space><elementName>:lastchild

```
<form id="frmLogin" method="post" action="/index.php/auth/validations">
<input id="txtUsername" type="text" name="txtUsername">
<input name="txtPassword" id="txtPassword" type="password">
<input name="Submit" class="button" id="btnLogin" value="LOGIN" type="submit">
</form>
```

Example :

To locate the submit button

Css: form#frmLogin input:last-child

10. Locating Elements by Prefix of the (startswith) Attribute Value

Syntax : elementName[attributeName^='prefixofthevalue']

<input id="txtUsername" type="text" name="txtUsername" class='userName' >

Examples :

Css: input[id^='txtUser']

Css: input[name^='txtUser']

11. Locating Elements by Suffix of the (endswith) Attribute Value

Value

Syntax : elementName[attributeName$='suffixofthevalue'] or *[attributeName$='suffixofthevalue']

<input id="txtUsername" type="text" name="txtUsername" class='userName' >

Examples :

Css: input[id$='Username']

Css: input[name$='name']

Css: *[id$='Username']

Css: *[name$='name']

12. Locating Elements containing part of the Attribute Value

Syntax : elementName[attributeName*='partofthevalue']

```
<input id="txtUsername" type="text" name="txtUsername" class='userName' >
```

Example :

Css: input[id*='User']

Css: input[name*='User']

13. Locating Next Sibling of a Known Element

Syntax : <CSSofelement> + siblingElement

```
<form id="frmLogin" method="post" action="/index.php/auth/validations">
<input id="txtUsername" type="text" name="txtUsername">
<input name="txtPassword" id="txtPassword" type="password">
<input name="Submit" class="button" id="btnLogin" value="LOGIN" type="submit">
</form>
```

Example :

To locate the password field

input#txtUsername + input

Quick CSS summary

CSS	Description
#ID	#ID Locate an element by ID
.className	Locate an element by class name
.className1.className2	Locate an element with with given classes
element[attribute='value']	Locate an element with an attribute and value
element[attribute1='value1'][attribute2='value2']	Locate an element with multiple attributes and values
parentElement > directChildElement	Locate direct child of the parent element
parentElement subChildElement	Locate an element inside another element(child at any level)
element[attribute^='prefixOfvalue']	Locating Elements by Prefix of the (startswith) Attribute
element[attribute$='suffixOfvalue']	Locating Elements by Suffix of the (endswith) Attribute
element[attribute*='partOfvalue']	Locating Elements containing part of the Attribute Value
CSSofParentElement elementName:nthoftype(N)	Locating Nth Child Elements by Element type
CSSofParentElement elementName:firstchild	Locating First Child Element by Element Name
CSS1 , CSS2	Combine the selectors with commas
CSS1 + CSS2	Locate immediately following sibling
CSS1 + * + CSS2	Locate following sibling with an intermediate sibling